

Assignment 2: Coding Basics

Kyle Falk

OVERVIEW

This exercise accompanies the lessons/labs in Environmental Data Analytics on coding basics.

Directions

1. Rename this file `<FirstLast>_A02_CodingBasics.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure to **answer the questions** in this assignment document.
5. When you have completed the assignment, **Knit** the text and code into a single PDF file.
6. After Knitting, submit the completed exercise (PDF file) to Canvas.

Basics, Part 1

1. Generate a sequence of numbers from one to 55, increasing by fives. Assign this sequence a name.
2. Compute the mean and median of this sequence.
3. Ask R to determine whether the mean is greater than the median.
4. Insert comments in your code to describe what you are doing.

```
#1.  
Sequence1 <-c(1,5,10,15,20,25,30,35,40,45,50,55) #Created term for Sequence1  
print(Sequence1)
```

```
## [1] 1 5 10 15 20 25 30 35 40 45 50 55
```

```
#2.  
the_mean <- mean(Sequence1) #created term to calculate mean of Sequence1 using mean function  
print(the_mean)
```

```
## [1] 27.58333
```

```
the_median <- median(Sequence1) #created term to calculate median of Sequence1 using median function  
print(the_median)
```

```
## [1] 27.5
```

```
#3.
meanvsmedian <- the_mean>the_median #created term to compare mean and median using greater than
print(meanvsmedian)
```

```
## [1] TRUE
```

Basics, Part 2

5. Create three vectors, each with four components, consisting of (a) student names, (b) test scores, and (c) whether they are on scholarship or not (TRUE or FALSE).
6. Label each vector with a comment on what type of vector it is.
7. Combine each of the vectors into a data frame. Assign the data frame an informative name.
8. Label the columns of your data frame with informative titles.

```
#5/6.
NameVector <- c("Kyle","LeBron","John","Damian") #Character vector
GradeVector <- c(100,45,60,95) #Number vector
ScholarshipVector <- c(TRUE,TRUE,FALSE,FALSE) #Logical value vector
```

```
#7/8.
is.data.frame(NameVector)
```

```
## [1] FALSE
```

```
is.data.frame(GradeVector)
```

```
## [1] FALSE
```

```
is.data.frame(ScholarshipVector)
```

```
## [1] FALSE
```

```
Correlation_of_Grades_and_Scholarships_at_Nicholas_School <- data.frame("Name"=NameVector,"Grade"=GradeVector,"Scholarship"=ScholarshipVector)
print(Correlation_of_Grades_and_Scholarships_at_Nicholas_School)
```

```
##      Name Grade Scholarship
## 1   Kyle   100         TRUE
## 2 LeBron    45         TRUE
## 3   John    60        FALSE
## 4 Damian    95        FALSE
```

9. QUESTION: How is this data frame different from a matrix?

Answer: A matrix contains 2 elements of the same type. This data frame has a characters, numbers, and logical values, so it cannot be a matrix.

10. Create a function with one input. In this function, use `if...else` to evaluate the value of the input: if it is greater than 50, print the word “Pass”; otherwise print the word “Fail”.
11. Create a second function that does the exact same thing as the previous one but uses `ifelse()` instead of `if...else`.
12. Run both functions using the value 52.5 as the input
13. Run both functions using the **vector** of student test scores you created as the input. (Only one will work properly...)

#10. Create a function using if...else

```
GradeComparison1 <- function(x){if(x>50){print("Pass")}
  else{print("Fail")}}
```

#11. Create a function using ifelse()

```
GradeComparison2 <- function(x){ifelse(x>50, "Pass", "Fail")}
```

#12a. Run the first function with the value 52.5

```
output1 <- GradeComparison1(52.5)
```

```
## [1] "Pass"
```

```
print(output1) #Pass
```

```
## [1] "Pass"
```

#12b. Run the second function with the value 52.5

```
output2 <- GradeComparison2(52.5)
print(output2) #Pass
```

```
## [1] "Pass"
```

#13a. Run the first function with the vector of test scores

```
#output3 <- GradeComparison1(GradeVector)
#print(output3)
```

#13b. Run the second function with the vector of test scores

```
output4 <- GradeComparison2(GradeVector)
print(output4)
```

```
## [1] "Pass" "Fail" "Pass" "Pass"
```

14. QUESTION: Which option of `if...else` vs. `ifelse` worked? Why? (Hint: search the web for “R vectorization”)

Answer: The second option, `ifelse` worked. This is because the function is vectorized, it is equipped to handle multiple conditions and applies operations to entire vectors simultaneously. The `if...else` function is not, meaning it applies to only one value at a time.

NOTE Before knitting, you'll need to comment out the call to the function in Q13 that does not work. (A document can't knit if the code it contains causes an error!)